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Review article

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# Optimizing antithrombotic therapy in patients with coexisting cardiovascular and gastrointestinal disease

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## Abstract

Balancing the safety and efficacy of antithrombotic agents in patients with gastrointestinal disorders is challenging because of the potential for interference with the absorption of antithrombotic drugs and for an increased risk of bleeding. In this Review, we address considerations for enteral antithrombotic therapy in patients with cardiovascular disease and gastrointestinal comorbidities. For those with gastrointestinal bleeding (GIB), we summarize a general scheme for risk stratification and clinical evidence on risk reduction approaches, such as limiting the use of concomitant medications that increase the risk of GIB and the potential utility of gastrointestinal protection strategies (such as proton pump inhibitors or histamine type 2 receptor antagonists). Furthermore, we summarize the best available evidence and potential gaps in our knowledge on tailoring antithrombotic therapy in patients with active or recent GIB and in those at high risk of GIB but without active or recent GIB. Finally, we review the recommendations provided by major medical societies, highlighting the crucial role of teamwork and multidisciplinary discussions to customize the antithrombotic regimen in patients with coexisting cardiovascular and gastrointestinal diseases.

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## Key points

- Balancing the safety and efficacy of antithrombotic therapy in patients with gastrointestinal disorders is challenging because of the potential for interference with the absorption of antithrombotic drugs or for an increased risk of bleeding.
- Various clinical states, including malabsorption syndromes, bariatric surgery, short-bowel syndrome or enteral tube feeding, can influence the absorption and bioavailability of oral antithrombotic agents.
- Bleeding events are an essential prognosticator in patients with cardiovascular diseases — at times as important as thrombotic events — and using antithrombotic agents in patients at high risk of gastrointestinal bleeding (GIB) is very challenging.
- Most of the existing models to predict the risk of bleeding in patients with coronary artery disease do not estimate the risk of GIB specifically.
- Identifying patients at high risk of GIB, modifying the bleeding risk by using gastroprotective agents, and determining the appropriate antithrombotic therapy regimen have crucial roles in preventing GIB.
- After an episode of acute GIB, determining the duration of antithrombotic therapy interruption and the regimen for re-initiation requires consideration of the balance between the bleeding severity and the risk of thrombotic events.

## Introduction

Balancing the efficacy and safety of antithrombotic therapy is delicate, especially in patients who have coexisting cardiovascular and gastrointestinal diseases, which can affect the pharmacokinetic and/or pharmacodynamic (PK/PD) properties of the drugs or modulate the risk of gastrointestinal bleeding (GIB). Clarity is needed about several aspects of managing antithrombotic therapy in these patients. First, various clinical states, such malabsorption syndromes, bariatric surgeries, short-bowel syndrome or enteral tube feeding can influence the absorption and bioavailability of oral antithrombotic agents[1,2]. Second, the use of antithrombotic agents in patients at high risk of GIB is nuanced. Bleeding events are an essential prognosticator in patients with cardiovascular disease and, at times, can be as important as thrombotic events[3–5]. Identifying patients at high risk of GIB, modifying the bleeding risk by using gastroprotective agents, and determining the appropriate antithrombotic therapy regimen have crucial roles in preventing GIB[6,7]. Third, after an episode of acute GIB, determining the duration of antithrombotic therapy interruption and the regimen for re-initiation requires consideration of the balance between the bleeding severity and the risk of thrombotic events[8,9].

In this Review, we summarize the evidence on the application of antithrombotic therapy in patients with coexisting cardiovascular and gastrointestinal diseases. We convened a group of experts in several related fields. At the evidence synthesis level, systematic searches of PubMed were performed to address a-priori-specified questions. For questions on efficacy or comparative effectiveness, randomized clinical trials (RCTs) and, when appropriate, pooled analyses of trials were prioritized (see Supplementary Fig. 1). Succinct management recommendations were shared either based on existing guidelines (if the guidelines provided clear recommendations) or based on the synthesized evidence in this Review (if the guidelines did not provide clear recommendations). If >15% of the authors had different perspectives from the originally drafted statement for a recommendation, a survey using the Delphi method was pursued, until ≥85% agreement was reached. The management of acute GIB[10–12] and bleeding or thrombotic events in patients with cirrhosis[13] are not covered in this Review.

## Effects of gastrointestinal disease on PK/PD

Several clinical conditions can affect the PK/PD of antithrombotic agents (see Supplementary Fig. 2). The most common changes are in the motility of the gastrointestinal system, gastrointestinal pH, the gastrointestinal absorptive surface area (due to diseases or surgery) and the route of drug administration. The oral route for medication intake might not be appropriate in patients with altered mental status, those with severe nausea or vomiting, or those with gastrointestinal motility problems or ileus that can interfere with the appropriate delivery of the drug to the intestinal tract. The choice of delivery route also depends on the PK/PD profile of the drug (see Supplementary Fig. 3). Very few RCTs have compared the PK/PD or clinical effectiveness of oral versus alternative routes of administration of antithrombotic agents (Fig. 1).

For antiplatelet therapy, non-enteric-coated aspirin can be crushed and administered via a nasogastric tube (or per rectum as a suppository) at the same dose and with the same expected efficacy as oral administration[14]. Intravenous aspirin is available in European countries for patients with acute coronary syndrome (ACS) who cannot take oral medications. For P2Y purinoceptor 12 ($P2Y_{12}$) inhibitors, pharmacokinetics data from healthy volunteers suggest that clopidogrel can be crushed and administered via a nasogastric tube with a faster onset of action and greater bioavailability than for whole tablets[15]. RCTs have demonstrated the superiority of crushed ticagrelor tablets versus whole pills in decreasing platelet reactivity in patients with ST-segment elevation myocardial infarction undergoing primary percutaneous coronary intervention (PCI)[16,17]. For prasugrel, in the CRUSH[17] ($n$ = 52) and COMPARE CRUSH[18] ($n$ = 727) RCTs, crushed prasugrel was superior to whole pills in terms of platelet inhibition and pharmacokinetic parameters in patients with ST-segment elevation myocardial infarction undergoing primary PCI. By contrast, in the FABOLUS-FASTER RCT[19], chewing the prasugrel tablets was not associated with higher platelet inhibition in a sub-randomization assessment of 42 patients with available light transmission aggregometry data. Moreover, in the largest of these trials (COMPARE-CRUSH[18]), no significant benefit was observed with crushed prasugrel tablets compared with whole pills in the primary surrogate outcomes of the percentage of patients reaching thrombolysis in myocardial infarction flow grade 3 in the infarct-related artery on initial angiography or achieving ≥70% ST-segment elevation resolution at 1 h after PCI. While crushing or chewing of $P2Y_{12}$ inhibitor pills might plausibly improve the PK/PD of these agents, such an effect has not been conclusively demonstrated and, more importantly, whether these PK/PD changes translate into a clinical benefit remains uncertain and requires further RCTs. In patients for whom oral intake of antiplatelet therapy is not feasible, intravenous cangrelor has been used successfully[20,21].

For oral anticoagulants (OACs), vitamin K antagonists can be administered via enteral access. Bioavailability can be unpredictable because of binding of the vitamin K antagonist to the enteral access tubes, necessitating close monitoring of the international normalized ratio[22,23]. For direct OACs (DOACs), dabigatran capsules should not be opened or crushed because the drug formulation has a coating applied

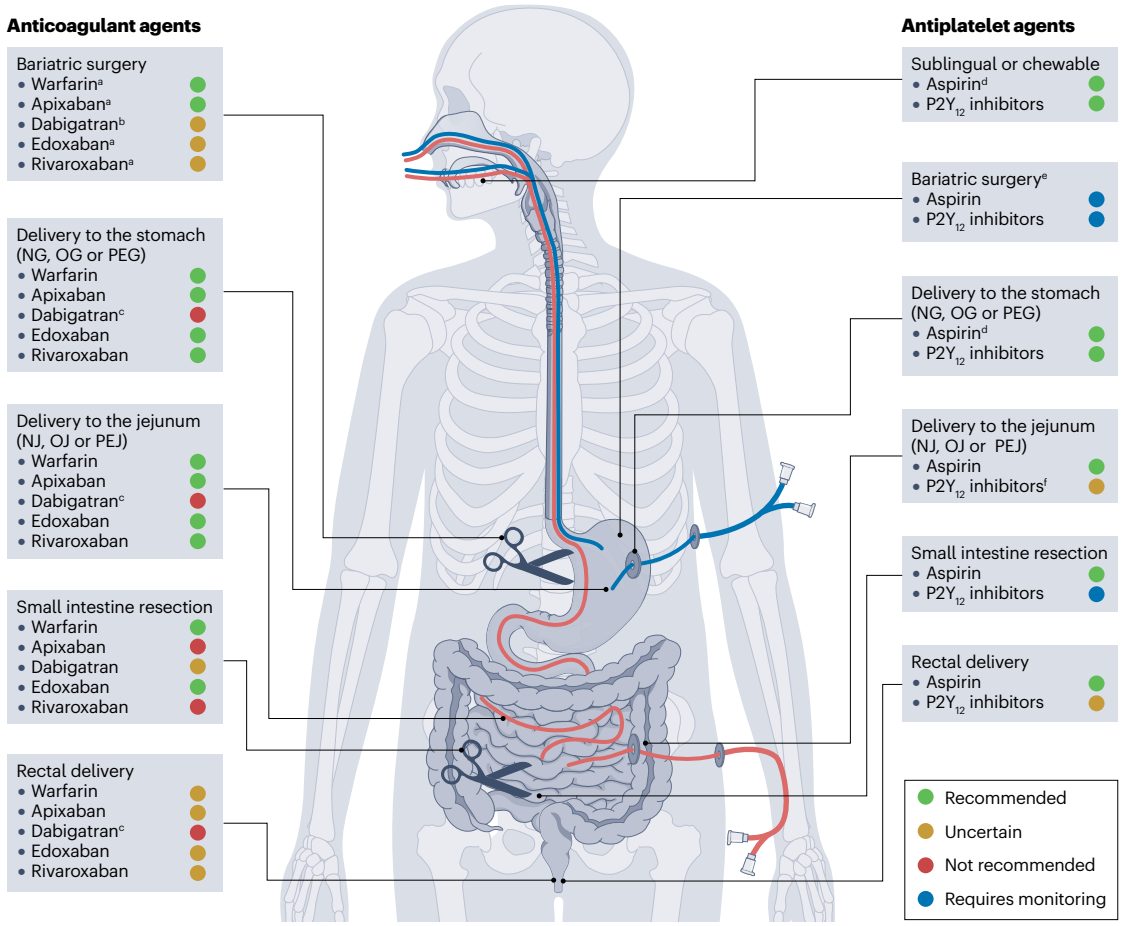


**Fig. 1 | Pharmacokinetics and pharmacodynamics of antithrombotic agents in patients with malabsorption, bariatric surgery or small-intestine resection or with delivery by different types of enteral tube.** Antithrombotic agents that can be crushed and used via tubes in patients who cannot take medications by the conventional oral route of administration are shown. [a]In 2021, the International Society on Thrombosis and Haemostasis[66] suggested not using direct oral anticoagulants (DOACs) for the treatment or prevention of venous thromboembolism in the acute setting after bariatric surgery, and suggested switching parenteral agents to a vitamin K antagonist. They advised that DOACs could be considered after at least 4 weeks of parenteral treatment, with monitoring of the DOAC trough level to ascertain absorption. However, based on limited more recent evidence, the effect of sleeve gastrectomy and Roux-en-Y gastric bypass on the pharmacokinetics and pharmacodynamics of apixaban is thought to be fairly small and most likely not clinically relevant. [b]Given that dabigatran is absorbed mostly in the stomach and duodenum, absorption might be decreased after bariatric surgery. [c]Dabigatran capsules should not be opened or crushed. [d]Non-enteric-coated formulations of aspirin. [e]After bariatric surgery, the absorption of aspirin and P2Y purinoceptor 12 ($P2Y_{12}$) inhibitors might be increased, necessitating monitoring for bleeding. [f]Given that $P2Y_{12}$ inhibitors are absorbed mostly in the proximal small bowel, their absorption might be decreased when delivering the drug into the distal parts of the small intestine. NG, nasogastric; NJ, nasojejunal; OG, orogastric, OJ orojejunal; PEG, percutaneous endoscopic gastrostomy; PEJ, percutaneous endoscopic jejunostomy.

onto a tartaric acid core to form tiny pellets to reduce variability in absorption[24]. Apixaban, edoxaban and rivaroxaban can be crushed and suspended in 60 ml of water (or apple juice or apple sauce)[25,26]. A rivaroxaban suspension also exists that can be administered via a gastric tube[27]. Given that rivaroxaban is absorbed primarily through the gastric mucosa, administration of rivaroxaban distal to the stomach is best avoided.

### Opioid-induced intestinal motility disorders

Opioids, including morphine (in doses as low as 5 mg) and fentanyl, delay gastric emptying and reduce intestinal motility, resulting in delayed $P2Y_{12}$ inhibitor absorption and antiplatelet response[28,29]. Fentanyl[30], intravenous acetaminophen[31] and lidocaine[32,33] have been studied as alternatives to morphine for pain control in patients with ACS, without conclusive evidence of clinical benefit (Supplementary Table 1). In high-risk patients undergoing PCI who received opiates, opioid receptor antagonists (methylnaltrexone and naloxone) did not improve platelet inhibition compared with placebo[34–36]. Concomitant administration of metoclopramide (a promotility agent) produces higher ticagrelor plasma concentrations and lower platelet reactivity[37,38]. Glycoprotein IIb/IIIa inhibitors have also been used to overcome the gap in platelet inhibition induced by opioid use[39,40]. However, RCTs of these agents focused on surrogate markers and were not powered for clinical outcomes. Based on the limited existing data,

the Canadian Cardiovascular Society discourages the routine use of opioids in patients presenting with ST-segment elevation myocardial infarction[41,42]. The European Society of Cardiology (ESC) points to the absence of a demonstrable increase in the risk of adverse clinical outcomes as a result of any interaction between morphine and antiplatelet agents in the setting of ACS[41,42].

### Absorption after gastrointestinal resection surgery

Gastrointestinal resection surgery can affect the absorption of antithrombotic agents. Aspirin absorption does not seem to be particularly affected in patients with gastrectomy or short-bowel syndrome. $P2Y_{12}$ inhibitors are absorbed mostly in the proximal small bowel; therefore, surgical procedures that manipulate the proximal small intestine, or the presence of short-bowel syndrome, can decrease their absorption[43]. Monitoring of platelet function in patients at high risk of thrombotic events might be reasonable until more data are available[44].

Vitamin K antagonists are absorbed in the stomach and proximal small intestine. However, given that the gut microbiota can affect the synthesis of menaquinone and/or the absorption of phylloquinone and menaquinone and, therefore, the response to the vitamin K antagonist[45], either hypo-responsiveness or hyper-responsiveness can occur in short-bowel syndrome, which necessitates close monitoring of the international normalized ratio[46,47]. Vitamin K antagonist bioavailability is unlikely to be affected by colectomy.

The bioavailability of DOACs is dependent on the specific site of absorption of the drug, together with the expression and distribution of cytochrome P450 3A4 (CYP3A4) through the intestine (especially in the jejunum). CYP3A4 expression is particularly important for apixaban and rivaroxaban (for which CYP3A4 is a primary pathway of metabolism), but less important for edoxaban (for which CYP3A4 is responsible for <4% of metabolism)[48]. Similarly, the distribution of P-glycoprotein 1 in the ileum and colon is important for the clearance of dabigatran and edoxaban, but less so for the clearance of apixaban and rivaroxaban. Dabigatran is mostly absorbed in the stomach and duodenum, and its absorption is impaired after gastrectomy or in short-bowel syndrome[49] but not after colectomy[50]. The absorption of rivaroxaban can be affected by gastrectomy but not by small-intestine resection procedures or colectomy[51,52]. Apixaban absorption occurs mainly in the small intestine and proximal large intestines; therefore, resection of the small intestine (but not gastrectomy) makes bioavailability unpredictable[53]. Edoxaban has pH-dependent solubility, and its disintegration occurs in the stomach. Edoxaban absorption can, therefore, be reduced after gastrectomy or gastric bypass but not after distal intestinal resection or colectomy[54] (see Supplementary Fig. 3).

### Bariatric surgery

Sleeve gastrectomy and Roux-en-Y gastric bypass account for >90% of contemporary bariatric surgical operations. In sleeve gastrectomy, nearly 80% of the stomach is removed without rerouting the gastrointestinal tract. In Roux-en-Y gastric bypass, a small gastric pouch is created by transection of the upper stomach. A connection between the proximal gastric pouch and the small bowel is created to establish continuity of the gastrointestinal tract. Therefore, a large portion of the mid and distal stomach, the entire duodenum and the proximal jejunum will not be in contact with ingested food and medications. While both procedures reduce the gastric luminal absorptive surface, the proximal small bowel is additionally excluded after Roux-en-Y gastric bypass[55,56]. These alterations can influence the PK/PD of antithrombotic agents[57,58]. Obesity might be associated with reduced effects of aspirin, and the surgery could restore full platelet inhibition[59]. Early after sleeve gastrectomy or Roux-en-Y gastric bypass, absorption of and responsiveness to aspirin can be increased[1,60]. Whether this increase in aspirin responsiveness translates into an excess risk of bleeding in the early period after surgery requires further investigation. Potential excess bleeding was reported with $P2Y_{12}$ inhibitors after Roux-en-Y gastric bypass[61]. Additional studies are required to understand the reasons for this increased risk of bleeding.

The appropriate postoperative dosing of anticoagulants after bariatric surgery remains uncertain considering the substantial weight loss and alterations to the gastrointestinal tract. For patients receiving warfarin, more frequent testing of the international normalized ratio is required in the first 3 months[62]. Only a few small studies have examined the PK/PD of DOACs after bariatric surgery. A study reported major changes in the bioavailability of dabigatran, edoxaban and rivaroxaban after bariatric surgery, and suggested postoperative monitoring of the drug level[63]. A systematic review highlighted the paucity of data on the use of DOACs in patients after bariatric surgery[64]. In an RCT of 272 patients undergoing sleeve gastrectomy who were assigned to 10 mg of rivaroxaban daily for either 7 days or 28 days, only one event of asymptomatic deep-vein thrombosis was observed (in the extended-duration prophylaxis group)[65]. Based on the limited evidence available, the effects of sleeve gastrectomy or Roux-en-Y gastric bypass on the PK/PD of apixaban are probably not clinically important[53]. Although the International Society on Thrombosis and Haemostasis advised against the use of DOACs in the first 4 weeks after Roux-en-Y gastric bypass[66], this recommendation predated the above-mentioned evidence. Checking plasma anti-factor Xa levels in these patients to detect low anticoagulant drug levels might be reasonable. In patients who are in the subtherapeutic range despite taking the DOAC regularly, switching to an alternative agent (such as low-molecular-weight heparins) for short-term management might be reasonable.

## Gastroprotection strategies

### Risk of GIB with antithrombotic therapy

Increasing the dose or the potency of antithrombotic therapy or the use of combination antithrombotic therapy will increase the risk of GIB associated with antiplatelet agents. The risk of GIB is also modulated by the concomitant use of NSAIDs or corticosteroids[67,68]. In patients receiving NSAIDs, the risk of GIB is highest with non-selective NSAIDs, although aspirin and cyclooxygenase 2-selective NSAIDs also confer a higher risk of GIB than placebo[69,70]. Patients at high risk of GIB were excluded from most trials of antithrombotic therapy. Therefore, much of our knowledge comes from observational studies[71].

In a meta-analysis of RCTs in patients treated with aspirin (75–325 mg daily) compared with inert control agents, the risk of major GIB was increased with aspirin use (OR 1.55, 95% CI 1.27–1.90)[72]. Insufficient evidence is available about the potential reduced risk of gastrointestinal ulcers or the maintenance of drug efficacy with enteric-coated compared with plain aspirin pills[73–75]. A post hoc analysis assessed the effectiveness and safety of enteric-coated compared with plain aspirin among the participants of a clinical trial that tested the dose of aspirin for secondary prevention. Among those who reported the aspirin formulation at baseline, no significant differences were found between enteric-coated and plain aspirin in terms of both effectiveness outcomes (adjusted HR 0.94, 95% CI 0.80–1.09, $P = 0.40$) or safety outcomes (adjusted HR 0.82, 95% CI 0.49–1.37, $P = 0.46$)[76]. The risk of GIB with aspirin seems to be dose-related, given that patients with ACS in another RCT who received 300–325 mg of aspirin per day had a

small but significant increase in major GIB during 30 days of follow-up compared with those who received 75–100 mg of aspirin per day (0.4% versus 0.2%; $P = 0.04$)[77]. The risk of GIB increased further when aspirin was combined with clopidogrel, compared with aspirin monotherapy (OR 1.86, 95% CI 1.49–2.31)[72].

Approximately half of major bleeding events associated with OAC therapy arise from the gastrointestinal tract[78]. The incidence of major GIB in RCTs of OACs for the treatment of venous thromboembolism (VTE) or for stroke prevention in atrial fibrillation (AF) was 0.5–1.6 and 0.8–1.9 events per 100 patient-years, respectively[79].

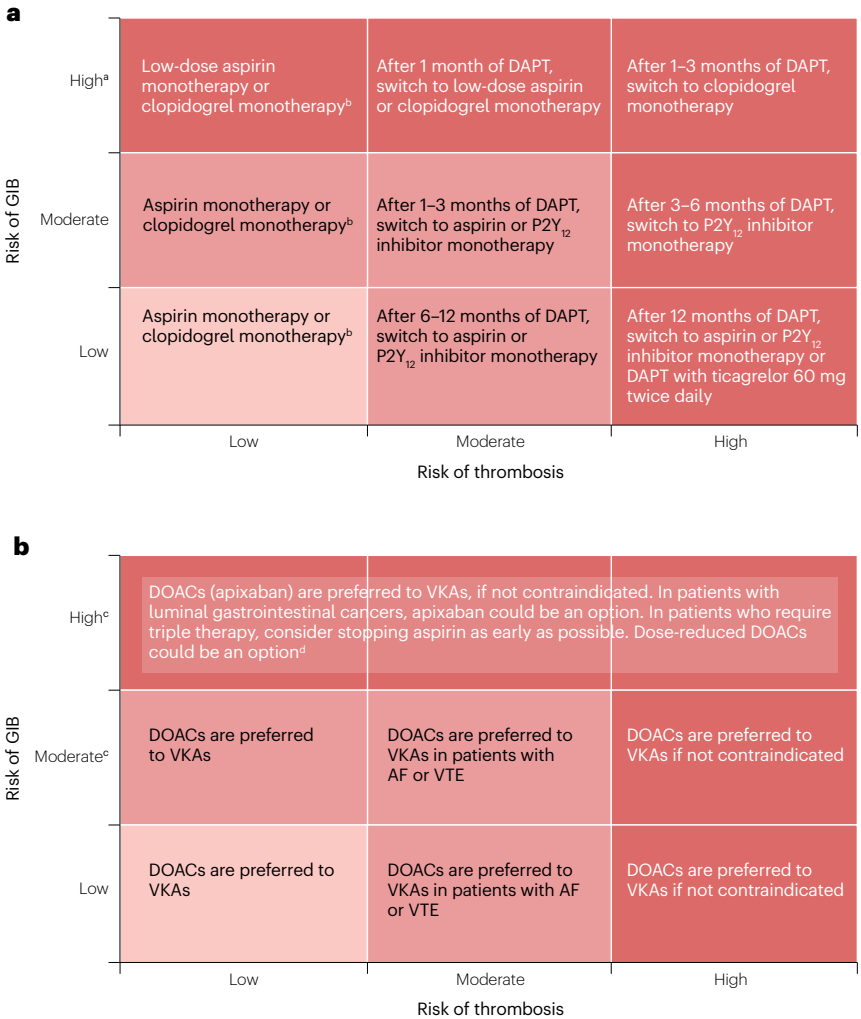


**Fig. 2 | Management of antithrombotic agents in patients at risk of GIB, stratified according to thrombotic risk and bleeding severity.** Management of antithrombotic agents in each patient must be individualized, and the summary in this figure is for general guidance only. Patients with a high risk of gastrointestinal bleeding (GIB) include those with luminal gastrointestinal cancers. Patients with a moderate risk of GIB include those with a history of peptic ulcer disease and those using NSAIDs. Patients with a low risk of GIB include those with a history of haemorrhoidal bleeding. **a**, Patients with a risk of thrombosis necessitating antiplatelet therapy. Patients at low risk of thrombosis include those receiving antiplatelet therapy for primary prevention or with asymptomatic atherosclerotic cardiovascular disease (ASCVD) without high-risk features. Patients at moderate risk of thrombosis include those with ASCVD without recent acute coronary syndrome (ACS) or percutaneous coronary intervention (PCI) and with high-risk features, those who have undergone elective PCI and those with medically managed non-ST-segment elevation ACS. Patients at high risk of thrombosis include those with ST-segment elevation myocardial infarction, ischaemic stroke, stent thrombosis or PCI in ACS. **b**, Patients with a risk of thrombosis necessitating anticoagulation. Patients at low risk of thrombosis include those with atrial fibrillation (AF) (a $CHA_2DS_2$-VASc score of 2–4), isolated distal deep-vein thrombosis, subsegmental pulmonary embolism (PE), or provoked venous thromboembolism (VTE) more than 3 months previously. Patients at moderate risk of thrombosis include those with intermediate–low risk PE, and those with mechanical valves other than in the mitral position, high-risk AF, or VTE 1–3 months previously. Patients at high risk of thrombosis include those with recent high-risk PE or those with intermediate–high risk PE, or those with a mechanical mitral valve, thrombotic antiphospholipid syndrome, or VTE in the past month. [a]In patients at high risk of GIB, clopidogrel is the P2Y purinoceptor 12 ($P2Y_{12}$) inhibitor of choice. [b]Clopidogrel is associated with a lower risk of bleeding than aspirin. [c]Left atrial appendage closure can be considered in patients with AF. [d]Dabigatran 110 mg twice daily could be an option in patients with AF (according to the European Medicines Agency), and rivaroxaban 15 mg daily could also be an option in patients with AF who require antiplatelet agents. DAPT, dual antiplatelet therapy; DOAC, direct oral anticoagulant; VKA, vitamin K antagonist.

**Table 1 | Recommendations on gastrointestinal protection strategies in patients receiving antithrombotic agents**

| Guideline document | Use of PPIs (or $H_2$ receptor antagonists) in patients with a high risk of bleeding receiving antithrombotic agents | Shortening the duration of triple therapy to reduce the risk of bleeding | Shortening the duration of dual antithrombotic therapy to reduce the risk of bleeding |
|---|---|---|---|
| 2020 ACC expert consensus decision pathway for anticoagulation and antiplatelet therapy in patients with AF or VTE undergoing PCI or with ASCVD[114] | PPIs (or $H_2$ receptor antagonists) are recommended in patients receiving two or more antithrombotic agents | Shortening the duration of triple therapy (maximum 1 month) is recommended to reduce the risk of bleeding | In stable patients who need long-term anticoagulation (due to AF or VTE), OAC alone (instead of an antiplatelet agent plus full-intensity OAC) is recommended after 1 year following PCI[a] |
| 2021 North American consensus on antithrombotic therapy in patients with AF undergoing PCI[115] | NA | Shortening the duration of triple therapy (maximum 1 month) is recommended to reduce the risk of bleeding | In stable patients with AF, OAC alone (instead of an antiplatelet agent plus full-intensity OAC) is recommended after 1 year following PCI[b] |
| 2023 ESC guidelines for the management of acute coronary syndromes[42] | PPIs in combination with DAPT are recommended in patients at high risk of GIB; PPIs are recommended in patients receiving a combination of antithrombotic agents or those receiving DAPT who are at high risk of GIB | Shortening the duration of triple therapy (up to 1 week) is recommended to reduce the risk of bleeding | Discontinuation of antiplatelet treatment in patients treated with an OAC is recommended after 12 months of treatment |

ACC, American College of Cardiology; AF, atrial fibrillation; ASCVD, atherosclerotic cardiovascular disease; DAPT, dual antiplatelet therapy; ESC, European Society of Cardiology; GIB, gastrointestinal bleeding; $H_2$, histamine type 2; NA, not addressed; OAC, oral anticoagulant; PCI, percutaneous coronary intervention; PPI, proton pump inhibitor; VTE, venous thromboembolism. [a]In patients at low risk of thrombosis and those at high risk of bleeding, it is reasonable to discontinue the $P2Y_{12}$ inhibitor therapy after 3 months for those presenting with stable ischaemic heart disease and after 6 months for those presenting with acute coronary syndrome. [b]In patients at low risk of thrombosis and those at high risk of bleeding, it is reasonable to discontinue the $P2Y_{12}$ inhibitor therapy 6 months after PCI.

RCTs in patients with AF showed that, compared with warfarin, the incidence of GIB was higher with dabigatran 150 mg twice daily (1.9 versus 1.3 events per 100 patient-years), edoxaban 60 mg daily (1.5 versus 1.3 events per 100 patient-years) and rivaroxaban (2.0 versus 1.2 events per 100 patient-years) but not with apixaban (0.8 versus 0.9 events per 100 patient-years)[79].

### Risk stratification for GIB

Various scores to estimate the risk of major bleeding (such as the PRECISE-DAPT[80] and the Academic Research Consortium for High Bleeding Risk definition[81]) have been used in patients receiving antiplatelet agents. However, these models do not specifically estimate the risk of GIB. The $(ABC)_2D$ score (age >65 years, history of anaemia, recent major bleeding, history of GIB, proton pump inhibitor (PPI) use, combined use of OACs and dual antiplatelet therapy (DAPT)) was developed as a risk score for the prediction of GIB in patients receiving antiplatelet agents[82] (see Supplementary Fig. 4).

We did not find a model to assess the risk of GIB in patients receiving OACs for AF. Of note, bleeding events might have different predictors depending on the site of bleeding[83]. In patients with VTE, the RIETE investigators identified male sex, age ≥70 years, initial VTE presentation as pulmonary embolism, active cancer, previous VTE, recent GIB, oesophageal varices, anaemia, abnormal prothrombin time, renal insufficiency and the use of corticosteroids as risk factors for major GIB[84]. The incidence of major GIB was 0.2% in low-risk individuals and 6.1% in high-risk individuals[84]. This model requires external validation.

### Pharmacotherapy for gastroprotection

In a meta-analysis of RCTs in patients taking aspirin (<325 mg daily), PPI use was associated with a reduced risk of GIB compared with the control group (OR 0.27, 95% CI 0.16–0.43)[85]. In the COGENT RCT[86] in 3,873 patients with ACS or undergoing PCI who received clopidogrel-based DAPT, a combination of clopidogrel (75 mg) and omeprazole (20 mg) compared with clopidogrel alone significantly reduced the 180-day rate of gastrointestinal events — a composite of overt or occult bleeding and symptomatic gastroduodenal ulcers or erosions, obstruction or perforation (HR 0.34, 95% CI 0.18–0.63, $P < 0.001$). Despite the interaction between omeprazole and clopidogrel, leading to delayed conversion of clopidogrel to its active metabolite, no clinically relevant increase in ischaemic events was observed. In patients receiving aspirin, PPIs provide better protection against GIB than histamine type 2 receptor antagonists. In an RCT in 60 patients with aspirin-related peptic ulcers or erosions, with or without a history of bleeding, more GIB events were reported among those who received the histamine type 2 receptor antagonist famotidine (40 mg twice daily) than among those who received the PPI pantoprazole (20 mg once daily) (7.7% versus 0%)[87]. In a separate RCT in patients receiving DAPT, esomeprazole 40 mg daily was associated with a lower rate of GIB than famotidine 20 mg daily (HR 0.095, 95% CI 0.005–0.504)[88]. In another small RCT in 270 high-risk patients with previous GIB, no significant difference in the rate of GIB with either rabeprazole or famotidine was observed (0.7% versus 3.1%; $P = 0.16$)[89]. The American College of Cardiology/American Heart Association guidelines recommend the use of PPIs in patients receiving DAPT who are at risk of GIB, including those receiving OAC, with advanced age, female sex, low body weight, chronic kidney disease, diabetes mellitus, anaemia or taking NSAIDs[90,91] (Fig. 2). Similarly, the ESC guidelines recommend the use of PPI with DAPT in patients at high risk of GIB (defined as those with a history of gastrointestinal ulcer or haemorrhage, anticoagulant therapy, chronic NSAID or corticosteroid use) or two or more of age ≥65 years, dyspepsia, gastro-oesophageal reflex disease, *Helicobacter pylori* infection or chronic alcohol use[42].

The COMPASS trial[92] compared the role of pantoprazole (40 mg daily) versus placebo among patients who were not excluded because of being at high risk of bleeding and who were receiving low-dose aspirin plus rivaroxaban 2.5 mg twice daily or aspirin monotherapy. The primary outcome was time to first upper gastrointestinal event, defined as a composite of overt bleeding, upper GIB, occult bleeding, symptomatic gastroduodenal ulcer or erosions, or upper gastrointestinal obstruction or perforation. The use of pantoprazole did not significantly reduce the risk of upper gastrointestinal events (the primary outcome) compared with placebo (HR 0.88, 95% CI 0.67–1.15) but was associated with a reduction in the risk of overt bleeding confirmed to be of gastroduodenal origin (HR 0.52, 95% CI 0.28–0.94, $P = 0.03$)[92].

### Risk of cardiac events with PPIs

The conversion of clopidogrel to its active metabolite requires transformation by hepatic cytochromes, mainly CYP2C19, which is inhibited by PPIs, particularly omeprazole and esomeprazole. Some observational studies have shown an association between PPI use and an increased risk of cardiovascular events in patients treated with clopidogrel[93–95]. However, in the COGENT RCT[86] in patients receiving clopidogrel-based DAPT, no significant difference was observed with omeprazole compared with placebo in the composite end point of cardiovascular death, non-fatal myocardial infarction, revascularization or stroke (HR 0.99, 95% CI 0.68–1.44). A subsequent meta-analysis of six RCTs found no significant difference in major adverse cardiovascular events with the use or non-use of PPIs (RR 0.99, 95% CI 0.76–1.28, $P = 0.93$)[96]. Meanwhile, in the 2023 ESC guidelines, the use of omeprazole or esomeprazole is not recommended in those receiving clopidogrel[97]. In patients receiving ticagrelor or prasugrel, PPI use is not associated with adverse cardiovascular outcomes[98] given that the metabolism of these drugs is not dependent on CYP2C19.

### Reducing the risk of GIB by cardiac procedures

Device strategies to abbreviate the course of the antithrombotic regimen among patients with AF include the left atrial appendage occluder or exclusion devices, such as percutaneous left atrial appendage closure. These devices have been shown in some RCTs to be non-inferior to anticoagulation for decreasing the incidence of stroke and systemic embolism[99–101]. Percutaneous left atrial appendage closure enables the de-escalation or cessation of anticoagulation, thereby reducing bleeding events. Data on the safety and efficacy of surgical closure devices are limited[102].

The design of coronary stents with biocompatible polymers or polymer-free stents[103,104] is among the strategies to allow an abbreviated duration and/or intensity of antiplatelet therapy, thereby reducing the risk of bleeding. The degradation of the polymer, which has been theorized to cause endothelial inflammation and stent failure, might allow the duration of DAPT after stenting to be safely abbreviated.

Insertion of an inferior vena cava filter is intended to reduce the risk of recurrent pulmonary embolism and is indicated in patients with acute proximal lower extremity deep-vein thrombosis or pulmonary embolism who have an absolute contraindication to anticoagulation. A systematic review of six RCTs and five prospective observational studies testing different indications for inferior vena cava filters found no significant difference in major bleeding events among patients treated and those not treated with an inferior vena cava filter (OR 0.90, 95% CI 0.58–1.38)[105]. These findings were verified in an updated analysis that included 13 studies[106].

## De-escalation of antithrombotic regimen

Choosing the best antithrombotic regimen for patients who need OAC together with one or two antiplatelet agents is challenging. Triple antithrombotic therapy in patients with AF or VTE who have had a recent ACS or who are undergoing PCI is associated with an increased risk of bleeding. RCTs suggest that, after a minimum duration of triple

**Table 2 | Recommendations on withholding and resuming antithrombotic agents in patients with acute GIB**

| Guideline document | Aspirin monotherapy in primary or secondary prophylaxis | Withholding aspirin and/or $P2Y_{12}$ inhibitors in patients receiving DAPT | OAC treatment |
|---|---|---|---|
| 2017 APAGE/APSDE guidelines on antithrombotic therapy management in patients undergoing endoscopy[176] | Withholding aspirin is recommended in patients with serious or life-threatening GIB; aspirin should be reinitiated after endoscopic haemostasis (within 3–5 days) | Withholding both agents simultaneously is not recommended in patients with coronary stents undergoing emergency endoscopy for non-variceal upper GIB[a]; $P2Y_{12}$ inhibitors should be reinitiated early (within 5 days) | OACs should be reinitiated after adequate haemostasis |
| 2017 ESC guidelines on antithrombotic therapy management after bleeding[142] | Withholding aspirin is recommended if endoscopy identifies a high risk of bleeding, but aspirin should be reinitiated as soon as bleeding is controlled (within 3 days); aspirin should be continued if endoscopy identifies a low risk of bleeding | DAPT should be continued if endoscopy identifies a low risk of bleeding. If endoscopy identifies a high risk of bleeding, the $P2Y_{12}$ inhibitor should be withheld. In patients at high or very high risk of thrombosis, aspirin should be continued. In patients at moderate risk of thrombosis, aspirin should be reinitiated within 3 days[b] | OACs should be reinitiated as soon as the thrombotic risk outweighs the bleeding risk (usually 1 week) |
| 2020 ACC expert consensus on bleeding management in patients receiving OACs[12] | NA | NA | OACs should be reinitiated after adequate haemostasis |
| 2021 ESGE guidelines for the diagnosis and management of upper GIB[147] | Withholding aspirin is recommended in primary prophylaxis settings, and aspirin should be reinitiated after careful re-evaluation for indication; aspirin should be continued in secondary prophylaxis settings but, if withheld, aspirin should be reinitiated as soon as bleeding is controlled (within 3–5 days) | Withholding the $P2Y_{12}$ inhibitor is recommended in patients with upper GIB, and it should be reinitiated early (within 5 days) | OACs should be reinitiated as soon as the thrombotic risk outweighs the bleeding risk (within 7 days) |
| 2022 ACG/CAG guidelines on antithrombotic therapy management during acute GIB and the periendoscopic period[11] | Withholding aspirin is not recommended in secondary prophylaxis settings and, if withheld, aspirin should be reinitiated after haemostasis is confirmed endoscopically | NA | NA |

ACC, American College of Cardiology; ACG, American College of Gastroenterology; APAGE, Asian Pacific Association of Gastroenterology; APSDE, Asian–Pacific Society for Digestive Endoscopy; CAG, Canadian Association of Gastroenterology; DAPT, dual antiplatelet therapy; ESC, European Society of Cardiology; ESGE, European Society of Gastrointestinal Endoscopy; GIB, gastrointestinal bleeding; NA, not addressed; OAC, oral anticoagulant; $P2Y_{12}$, P2Y purinoceptor 12. [a]In patients receiving proton pump inhibitor infusion and DAPT with aspirin and clopidogrel, continuing aspirin and withholding clopidogrel is recommended. [b]The second antiplatelet agent should be reinstated on an individual basis.

**Table 3 | Efficacy and bleeding outcomes in RCTs of antiplatelet regimens in patients with ASCVD**

| Trial | Treatment groups | Sample size | Population | Follow-up | Primary efficacy outcome | Safety outcomes | |
|---|---|---|---|---|---|---|---|
| | | | | | | Major bleeding | GIB |
| COMPASS (Japanese patients)[177] | Clopidogrel 75 mg daily versus clopidogrel 50 mg daily | 1,110 | Japanese patients aged 20–74 years with ischaemic stroke; excluded patients at high risk of bleeding, receiving OAC, or with malignancy, cardiac disease, or serious renal or severe hepatic dysfunction | 52 weeks | Vascular events: HR 1.31, 95% CI 0.68–2.51 | Serious bleeding adverse events: HR 1.24, 95% CI 0.49–3.14, *P* = 0.65 | Number of GIB events: 14/552 (2.5%) versus 12/558 (2.2%) |
| PLATO[126,178] | Ticagrelor 90 mg twice daily versus clopidogrel 75 mg daily in patients taking aspirin 75–100 mg daily | 18,624 | Adult patients with ACS; excluded patients who received fibrinolytic therapy or receiving OAC | 1 year | A composite of death from vascular causes, MI or stroke: HR 0.84, 95% CI 0.77–0.92 | Major bleeding: HR 1.04, 95% CI 0.95–1.13, *P* = 0.43 | Number of major GIB events: 124/9,235 (1.3%) versus 94/9,186 (1.0%) |
| TRITON-TIMI 38 (ref. 129) | Prasugrel 10 mg daily versus clopidogrel 75 mg daily | 13,608 | Adult patients with ACS with scheduled PCI, excluded patients at high risk of bleeding or had taken any thienopyridine in past 5 days | 14.5 months | Composite of cardiovascular death, non-fatal MI or stroke: HR 0.81, 95% CI 0.73–0.90 | Major TIMI bleeding: HR 1.32, 95% CI 1.03–1.68, *P* = 0.03 | GIB, including TIMI major and minor bleeding events: HR 1.32, 95% CI 0.99–1.77, *P* = 0.05 |

Only RCTs in which GIB was reported are included. All trials were double-blinded. ACS, acute coronary syndrome; ASCVD, atherosclerotic cardiovascular disease; GIB, gastrointestinal bleeding; MI, myocardial infarction; OAC, oral anticoagulant; PCI, percutaneous coronary intervention; RCT, randomized controlled trial; TIMI, thrombolysis in myocardial infarction.

therapy (commonly 1 week to 1 month)[107], de-escalation to an anticoagulant agent plus a $P2Y_{12}$ inhibitor (commonly clopidogrel) is associated with a lower risk of bleeding, compared with ongoing triple therapy, without a significant difference in ischaemic events[108–113]. These data have been incorporated into some clinical guidelines[114,115] (Tables 1,2).

Dual antithrombotic therapy should also be prescribed for a limited period, if possible. Whether antiplatelet therapy should be continued in patients receiving long-term anticoagulation should be addressed by a thrombosis specialist. Evidence suggests that many patients taking anticoagulants can safely stop antiplatelet therapy. In patients with AF and stable atherosclerotic cardiovascular disease (no ACS or PCI in the previous year), anticoagulation monotherapy was associated with a lower risk of overall bleeding than an anticoagulant plus an antiplatelet agent, with no observed increase in the risk of ischaemic events[116–120]. In the AFIRE RCT[121], patients with AF and stable atherosclerotic cardiovascular disease (including those with PCI >1 year previously) were randomly assigned to receive rivaroxaban or rivaroxaban plus single antiplatelet therapy. Rivaroxaban monotherapy was non-inferior to combination therapy in terms of efficacy (a composite of stroke, systemic embolism, myocardial infarction, unstable angina requiring revascularization or death from any cause) (HR 0.72, 95% CI 0.55–0.95, $P < 0.001$ for non-inferiority) and superior for safety (major bleeding) (HR 0.59, 95% CI, 0.39–0.89, $P = 0.01$)[121]. A similar hypothesis is being tested in patients with acute VTE[122].

## Patients at high risk of GIB

### Antiplatelet therapy

In the CAPRIE RCT[123,124] involving patients at risk of ischaemic events, the risk of upper GIB was higher with 325 mg of aspirin daily than with 75 mg of clopidogrel daily (2.7% versus 2.0%; $P = 0.05$), as was the rate of hospitalization for GIB (1.1% versus 0.7%; $P = 0.012$). In a meta-analysis of 22 RCTs on primary and secondary prevention of cardiovascular disease, aspirin (75–325 mg per day) increased the risk of GIB compared with placebo (RR 2.07, 95% CI 1.61–2.66)[125]. In a patient-level meta-analysis of RCTs in patients with established coronary artery disease, $P2Y_{12}$ inhibitor monotherapy (ticagrelor 90 mg twice daily or clopidogrel 75 mg daily) was associated with a lower risk of any GIB (HR 0.75, 95% CI 0.57–0.97, $P = 0.027$) and ischaemic events (HR 0.88, 95% CI 0.79–0.97, $P = 0.012$) compared with aspirin monotherapy (75–325 mg daily)[119]. The lower risk of GIB was probably driven by clopidogrel, based on the results of major landmark trials[126,127]. In the EUCLID RCT[128], which compared ticagrelor with clopidogrel as monotherapy in patients with symptomatic peripheral artery disease, the risk of major bleeding was not significantly different between the two groups. Overall, 32 out of 6,932 patients assigned to clopidogrel and 17 out of 6,910 patients assigned to ticagrelor had an upper GIB event[128]. In the PLATO RCT[126], which compared ticagrelor with clopidogrel as a part of DAPT in patients with ACS, drug discontinuation due to non-procedural bleeding was related to GIB in 0.7% of patients receiving ticagrelor and 0.3% patients receiving clopidogrel. The cumulative incidence of GIB was also higher among those receiving ticagrelor than among those receiving clopidogrel (2.3% versus 1.9%)[126]. However, the case fatality rate was higher among patients with GIB who were assigned to clopidogrel than among those assigned to ticagrelor (2.8% versus 0%)[126]. In the TRITON-TIMI 38 trial[129], which compared prasugrel with clopidogrel as a part of DAPT in patients with ACS with scheduled PCI, the risk of GIB was higher among patients receiving prasugrel 10 mg daily than among those receiving clopidogrel 75 mg daily (HR 1.32, 95% CI 0.99–1.77, $P = 0.05$) (Table 3). In a subgroup analysis of RCTs in patients receiving third-generation $P2Y_{12}$ inhibitors (including trials of their use as monotherapy or as DAPT in combination with aspirin), the risk of GIB was higher among those receiving prasugrel (RR 1.40, 95% CI 1.10–1.77) than among those receiving ticagrelor (RR 1.15, 95% CI 0.94–1.39), each compared with clopidogrel[130]. Despite the limited evidence, the available data suggest that clopidogrel might be associated with a lower risk of GIB than either aspirin or prasugrel in patients with stable ischaemic risk[119,131,132]. Although clopidogrel might be associated with fewer GIB events than ticagrelor, the data are not conclusive, and there was concern about a higher case fatality rate with clopidogrel than with ticagrelor in one study. Therefore, no conclusions can be reached

on the comparison between these two drugs. Future analyses from the PANTHER collaborative[119], a patient-level meta-analysis of RCTs comparing $P2Y_{12}$ inhibitor monotherapy and aspirin monotherapy for the prevention of cardiovascular events in patients with established coronary artery disease, will be informative.

### Anticoagulant therapy

There is a dearth of high-quality studies into the use of anticoagulant agents in patients at high risk of GIB who have not had a recent GIB event. In patients with AF, based on the results of landmark trials, edoxaban 60 mg daily (HR 1.23, 95% CI 1.02–1.50, $P = 0.03$), dabigatran 150 mg twice daily (HR 1.50, 95% CI 1.19–1.89, $P < 0.001$) and rivaroxaban 20 mg daily (HR 1.42, 95% CI 1.22–1.66, $P < 0.001$) were associated with a higher risk of GIB than warfarin[133–135]. Conversely, edoxaban 30 mg daily was associated with a lower risk of GIB than warfarin (HR 0.67, 95% CI 0.57–0.83, $P < 0.001$)[133], whereas the risk of GIB with dabigatran 110 mg twice daily was not significantly different from that with warfarin (HR 1.10, 95% CI 0.86–1.41, $P = 0.43$)[135] (Table 4). Low-intensity edoxaban and dabigatran have not been approved by the FDA for stroke prevention. In patients with VTE, GIB events were more frequent with dabigatran than with warfarin (4.2% versus 2.8% in the RE-COVER I RCT[136] and 3.7% versus 2.6% in the RE-COVER II RCT[137]), whereas GIB events were less frequent with apixaban than with warfarin (0.3% versus 0.7% in the AMPLIFY trial[138]) (Fig. 2; Table 4). In patients with cancer and VTE, GIB events were numerically (but not significantly) more frequent with rivaroxaban than with dalteparin

**Table 4 | Efficacy and bleeding outcomes in RCTs of DOACs versus warfarin in patients with AF**

| Trial | Treatment groups | Sample size | Study population | Follow-up (years) | Primary efficacy outcomes | Safety outcomes | |
|---|---|---|---|---|---|---|---|
| | | | | | | Major bleeding | GIB event rate (% per year) |
| ARISTOTLE[179] | Apixaban (5 mg twice daily or 2.5 mg twice daily if dose-adjustment criteria were fulfilled) versus warfarin | 18,201 | Adult patients with AF and $CHADS_2$ score ≥1; excluded patients with moderate-to-severe MS, recent stroke, hepatic or severe renal insufficiency, receiving OAC for other reasons, at high risk of bleeding, taking DAPT or aspirin >165 mg per day | 1.8 | Composite of stroke or systemic embolism: HR 0.79, 95% CI 0.66–0.95, $P_{NI} < 0.001$ | ISTH major bleeding: HR 0.69, 95% CI 0.60–0.80, $P < 0.001$ | GIB: 105/9,088 (0.76%/year) versus 119/9,052 (0.86%/year) (HR 0.89, 95% CI 0.70–1.15, $P = 0.37$) |
| ENGAGE AF-TIMI 48 (ref. 133) | Edoxaban (30 mg daily) versus warfarin | 21,105 | Age ≥21 years with AF and $CHADS_2$ score ≥2; excluded patients with moderate-to-severe MS, recent stroke, hepatic or severe renal insufficiency, receiving OAC for other reasons, at high risk of bleeding, taking DAPT or aspirin >165 mg per day, ACS, PCI | 2.8 | Composite of stroke or systemic embolism: HR 1.07, 97.5% CI 0.87–1.31, $P_{NI} = 0.005$ | Total major bleeding events: HR 0.47, 95% CI 0.41–0.55, $P < 0.001$ | Total upper and lower GIB events: 129/7,002 (0.82%/year) versus 190/7,012 (1.23%/year) (HR 0.67, 95% CI 0.53–0.83, $P < 0.001$) |
| | Edoxaban (60 mg daily) versus warfarin | | | | HR 0.79, 97.5% CI 0.63–0.99, $P_{NI} < 0.001$ | HR 0.80, 95% CI 0.71–0.91, $P < 0.001$ | 232/7,012 (1.51%/year) versus 190/7,012 (1.23%/year) (HR 1.23, 95% CI 1.02–1.50, $P = 0.03$) |
| RE-LY[135] | Dabigatran (110 mg twice daily) versus warfarin | 18,113 | Adult patients with AF at high risk of thrombosis; excluded patients with valvular heart disease, recent stroke, hepatic or severe renal insufficiency, receiving OAC for other reasons, at high risk of bleeding | 2 | Composite of stroke and systemic embolism: RR 0.91, 95% CI 0.74–1.11, $P_{NI} < 0.001$ | Total major bleeding events: RR 0.80, 95% CI 0.69–0.91, $P = 0.003$ | GIB: 133/6,015 (1.12%/year) versus 120/6,022 (1.02%/year) (RR 1.10, 95% CI 0.86–1.41, $P = 0.43$) |
| | Dabigatran (150 mg twice daily) versus warfarin | | | | RR 0.66, 95% CI 0.53–0.82, $P < 0.001$ | RR 0.93, 95% CI 0.81–1.07, $P = 0.31$ | 182/6,076 (1.51%/year) versus 120/6,022 (1.02%/year) (RR 1.50, 95% CI 1.19–1.89, $P < 0.001$) |
| | Dabigatran (150 mg twice daily) versus dabigatran (110 mg twice daily) | | | | RR 0.73, 95% CI 0.58–0.91, $P = 0.005$ | RR 1.16, 95% CI 1.00–1.34, $P = 0.052$ | 182/6,076 (1.51%/year) versus 133/6,015 (1.12%/year) (RR 1.36, 95% CI 1.09–1.70, $P = 0.007$) |
| ROCKET AF[134] | Rivaroxaban (20 mg daily) versus warfarin | 14,264 | Adult patients with non-valvular AF and $CHADS_2$ score ≥2; excluded patients with significant MS, recent stroke, hepatic or severe renal insufficiency, receiving OAC for other reasons, at high risk of bleeding, taking DAPT or aspirin >100 mg per day | 1.9 | Composite of stroke or systemic embolism: HR 0.79, 95% CI 0.66–0.96, $P_{NI} < 0.001$ | Total major bleeding events: HR 1.04, 95% CI 0.90–1.20, $P = 0.58$ | Major or non-major GIB[a]: 394/7,111 (3.61%/year) versus 290/7,125 (2.60%/year) (HR 1.42, 95% CI 1.22–1.66, $P < 0.0001$) |

All trials were double-blinded. In the warfarin groups, the target international normalized ratio was 2–3. ACS, acute coronary syndrome; AF, atrial fibrillation; DAPT, dual antiplatelet therapy; DOAC, direct oral anticoagulant; GIB, gastrointestinal bleeding; ISTH, International Society on Thrombosis and Haemostasis; MS, mitral stenosis; OAC, oral anticoagulant; PCI, percutaneous coronary intervention; $P_{NI}$, $P$ value for non-inferiority; RCT, randomized controlled trial. [a]Event rate per 100 patient-years.

**Table 5 | Efficacy and bleeding outcomes in RCTs of DOACs versus other anticoagulants in patients with VTE**

| Trial | Blinding | Comparison groups | Sample size | Study population | Follow-up (months) | Primary efficacy outcome | Safety outcomes | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Major bleeding | GIB event rate |
| **DOACs versus warfarin in patients with VTE** | | | | | | | | |
| AMPLIFY[138] | Open label | Apixaban (10 mg twice daily for 7 days followed by 5 mg twice daily) versus warfarin | 5,395 | Adult patients with acute symptomatic VTE; excluded patients with hepatic or renal insufficiency, at high risk of bleeding, taking DAPT or aspirin >165 mg per day | 6 | Symptomatic recurrent VTE or related death: HR 0.84, 95% CI 0.60–1.18, $P_{NI}$ < 0.001 | Total major bleeding events: HR 0.31, 95% CI 0.17–0.55, $P$ < 0.001 | 7/2,691 (0.3%) versus 18/2,704 (0.7%) |
| RE-COVER I[136] | Double blind | Dabigatran (150 mg twice daily) versus warfarin | 2,539 | Adult patients with acute symptomatic VTE; excluded patients with hepatic or renal insufficiency, PE haemodynamic instability or symptoms >14 days, at high risk of bleeding, need for antiplatelet therapy (except aspirin ≤100 mg) | 6 | Symptomatic recurrent VTE or related death: HR 1.10, 95% CI 0.65–1.84, $P_{NI}$ < 0.001 | Total major bleeding events: HR 0.82, 95% CI 0.45–1.48 | 53/1,274 (4.16%) versus 35/1,265 (2.77%) |
| RE-COVER II[137] | Double blind | Dabigatran (150 mg twice daily) versus warfarin | 2,589 | Adult patients with acute symptomatic VTE; excluded patients with hepatic or renal insufficiency, PE haemodynamic instability or symptoms >14 days, at high risk of bleeding, need for antiplatelet therapy (except aspirin ≤100 mg) | 6 | Symptomatic recurrent VTE or related death: HR 1.08, 95% CI 0.64–1.80, $P_{NI}$ < 0.001 | Total major bleeding events: HR 0.69, 95% CI 0.36–1.32 | 48/1,279 (3.75%) versus 33/1,289 (2.56%) |
| **DOACs versus LMWH in patients with cancer and VTE** | | | | | | | | |
| Caravaggio[141] | Open label | Apixaban (10 mg twice daily for 7 days, then 5 mg twice daily for a total of 6 months) versus dalteparin (200 IU/kg daily for 1 month, then 150 IU/kg daily for 2–6 months) | 1,170 | Adult patients with cancer and confirmed VTE; excluded patients with hepatic or renal insufficiency, receiving anticoagulant therapy, at high risk of bleeding | 7 | VTE recurrence: HR 0.63 95% CI 0.37–1.07, $P_{NI}$ < 0.001 | Total major bleeding events: HR 0.82, 95% CI 0.40–1.69, $P$ = 0.60 | 11/576 (1.91%) versus 10/579 (1.73%) (HR 1.05, 95% CI 0.44–2.50) |
| Hokusai[140] | Open label | Edoxaban (60 mg once daily after ≥5 days of parenteral therapy) versus dalteparin (200 IU/kg daily for 1 month, then 150 IU/kg daily for 6–12 months) | 1,050 | Adult patients with cancer and confirmed VTE; excluded patients with hepatic or renal insufficiency, receiving anticoagulant therapy, need for antiplatelet therapy (other than low-dose aspirin monotherapy), at high risk of bleeding, history of HIT | 12 | Composite of recurrent VTE or major bleeding: HR 0.97, 95% CI 0.70–1.36, $P_{NI}$ = 0.006 | Major bleeding: HR 1.77, 95% CI 1.03–3.04, $P$ = 0.04 | 20/522 (3.8%) versus 6/524 (1.1%) |
| SELECT-D[139] | Open label | Rivaroxaban (15 mg twice daily for 3 weeks, then 20 mg once daily for a total of 6 months) versus dalteparin (200 IU/kg daily for 1 month, then 150 IU/kg daily for 2–6 months) | 406 | Adult patients with active cancer and confirmed VTE, and body weight >40 kg; excluded patients with hepatic insufficiency, history of VTE, receiving anticoagulant therapy, need for antiplatelet therapy, at high risk of bleeding | 24 | VTE recurrence: HR 0.43, 95% CI 0.19–0.99 | Cumulative rate of major bleeding over 6 months: HR 1.83, 95% CI 0.68–4.96 | 9/203 (4.43%) versus 3/203 (1.48%) |

Only RCTs in which GIB was reported are included. In the warfarin groups, the target international normalized ratio was 2–3. DAPT, dual antiplatelet therapy; DOAC, direct oral anticoagulant; GIB, gastrointestinal bleeding; HIT, heparin-induced thrombocytopenia; IU, international units; LMWH, low-molecular-weight heparin; $P_{NI}$, $P$ value for non-inferiority; PE, pulmonary embolism; RCT, randomized controlled trial; VTE, venous thromboembolism.

## Box 1

# Key points for clinical practice

**Pharmacokinetic and pharmacodynamic properties of antithrombotic agents important for patients with gastrointestinal conditions**

- Chewable aspirin and oral P2Y purinoceptor 12 ($P2Y_{12}$) inhibitors can be crushed and administered via a gastric tube.
- Warfarin can be crushed and administered through a gastric tube. Close monitoring of the international normalized ratio (INR) is recommended.
- Apixaban, edoxaban and rivaroxaban can be crushed and administered via a gastric tube; however, dabigatran should not be crushed.
- The use of opioids in patients with ST-segment elevation myocardial infarction might delay the absorption of $P2Y_{12}$ inhibitors taken orally and the antiplatelet response. The effect on clinical outcomes is uncertain, with the limited available clinical data not demonstrating an increase in the risk of adverse clinical outcomes as a result of any interaction between morphine and antiplatelet agents in the setting of acute coronary syndrome.
- The bioavailability of aspirin and $P2Y_{12}$ inhibitors is affected by gastrectomy and short-bowel syndrome. Some investigators recommend platelet function monitoring.
- For warfarin use, close INR monitoring is needed after gastrectomy or in short-bowel syndrome.
- Apixaban is the preferred oral anticoagulant (OAC) after gastrectomy.
- Rivaroxaban and edoxaban are the preferred OACs in short-bowel syndrome.
- The bioavailability of antithrombotic agents is not affected by colectomy.
- Evidence for the benefits of using rivaroxaban or apixaban in patients undergoing bariatric surgery is limited but growing. INR monitoring is needed in users of vitamin K antagonists.

**Risk of gastrointestinal bleeding**

- Non-selective NSAIDs confer a higher risk of gastrointestinal bleeding (GIB) than selective NSAIDs.
- The risk of GIB is higher with prasugrel or ticagrelor than with clopidogrel.
- Full-intensity apixaban might confer a lower risk of GIB than other full-intensity OACs.

**Risk stratification**

- Most of the existing models for predicting the risk of bleeding in patients with coronary artery disease do not specifically estimate the risk of GIB. The $(ABC)_2D$ score predicts the risk of GIB in patients treated with antiplatelet therapy.
- No externally validated prediction models for GIB exist for use in patients with atrial fibrillation treated with OACs. The RIETE investigators proposed male sex, age >70 years, presentation as pulmonary embolism, active cancer, previous venous thromboembolism, recent GIB, oesophageal varices, anaemia, abnormal prothrombin time, renal insufficiency and the use of corticosteroids as risk factors for major GIB. This model requires external validation.

**Pharmacological gastrointestinal-protection strategies**

- Proton pump inhibitors (PPIs) might decrease the risk of GIB in patients receiving anticoagulant agents.
- PPIs decrease the risk of GIB in patients receiving antiplatelet agents.
- PPIs do not affect the bioavailability of prasugrel or ticagrelor.
- The interaction between clopidogrel and the PPIs omeprazole and esomeprazole seems not to be clinically relevant. However, the use of an alternative PPI (such as pantoprazole) is reasonable in patients taking clopidogrel.

**Antithrombotic therapy in patients at high risk of GIB**

- Clopidogrel might be the preferred $P2Y_{12}$ inhibitor in patients at high risk of GIB.
- Despite limited data, apixaban could be considered the OAC of choice in patients at high risk of GIB.
- For the long-term secondary prevention of coronary events, single antiplatelet therapy with clopidogrel or ticagrelor could be preferred to aspirin.

**Antithrombotic therapy in patients with active GIB**

- In patients with major active GIB, treatment with antithrombotic agents should be interrupted. The prompt use of specific reversal agents (if available) is suggested in patients needing urgent surgery or invasive procedures or who have life-threatening bleeding.
- In patients with definite indications for dual antiplatelet therapy (DAPT) (such as recent coronary stenting) and upper GIB, continuation of DAPT is suggested in those with low-risk endoscopic features. In patients with high-risk endoscopic features, the short-term interruption of antiplatelet therapy based on bleeding severity and thrombotic risk is suggested, with at least one drug (usually aspirin) being reintroduced early, and the second drug being reintroduced 5 days after achieving haemostasis.

**Antithrombotic therapy in patients with recent GIB**

- Delayed resumption of antithrombotic therapies after haemostasis could be harmful.
- Resumption of aspirin as a secondary prevention strategy in patients at high cardiovascular risk can be considered on the same day in patients with endoscopically confirmed haemostasis or after stabilization in those with diverticular haemorrhage.
- Anticoagulation resumption within 7 days after stabilization of bleeding is suggested in those with lower GIB or upper GIB and high thromboembolic risk.
- Anticoagulation resumption within 7–14 days after stabilization of bleeding is suggested in patients with upper GIB and low thromboembolic risk.

(4.4% versus 1.5% in the SELECT-D RCT[139]) and with edoxaban than with dalteparin (3.8% versus 1.1% in the Hokusai VTE Cancer RCT[140]). The number of GIB events was not significantly different between patients who received apixaban and those who received dalteparin (1.9% versus 1.7%)[141] (Table 5).

## Patients with active GIB

Despite the baseline risk of thrombosis in patients with cardiovascular diseases, antithrombotic therapy is commonly paused during periods of active, clinically relevant GIB (Box 1). Based on the clinical context, reversal agents can be used if indicated (Supplementary Table 2).

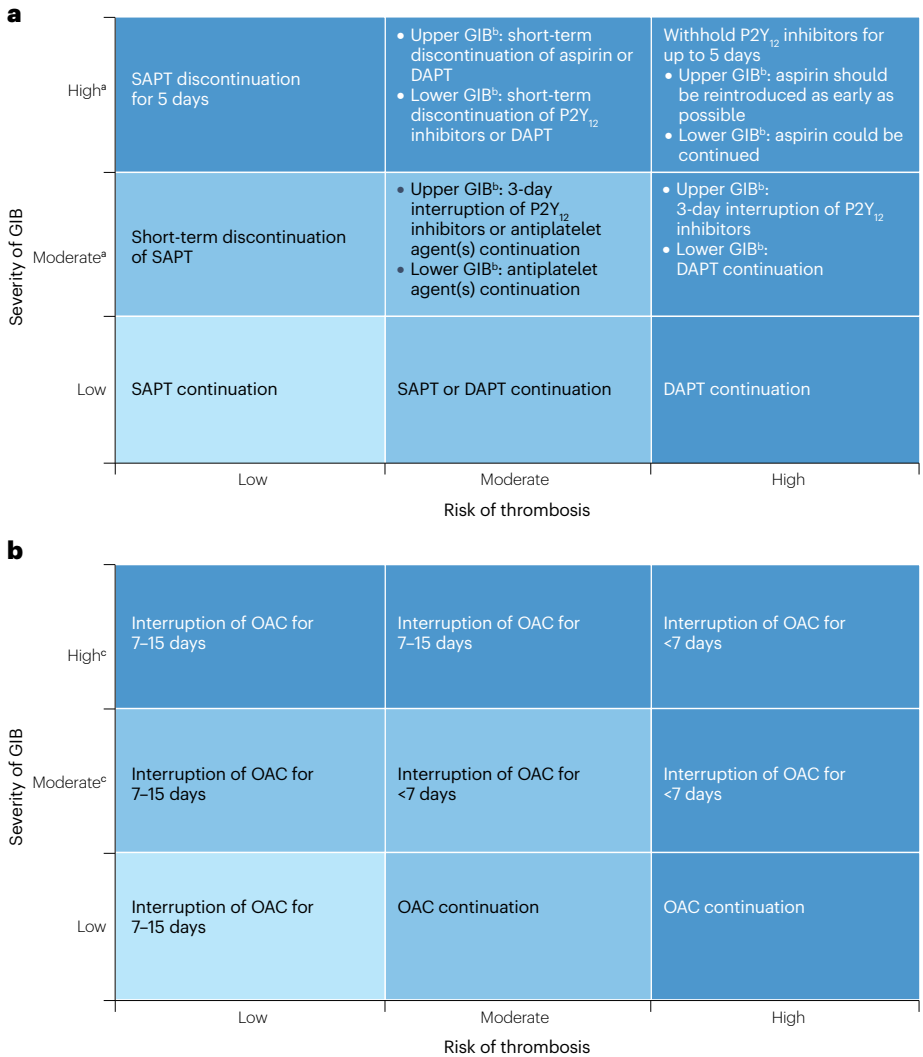


**Fig. 3 | Management of antithrombotic agents in patients with recent GIB, stratified according to thrombotic risk and bleeding severity.** Patients with high-risk recent gastrointestinal bleeding (GIB) include those with haemorrhagic shock or major GIB. Patients with moderate-risk recent GIB include those with clinically relevant non-major GIB. Patients with low-risk recent GIB include those with a small haemorrhoidal bleed or a Mallory–Weiss tear. **a**, Patients with a risk of thrombosis necessitating antiplatelet therapy. Patients at low risk of thrombosis include those with stable atherosclerotic cardiovascular disease (ASCVD) without high-risk features. Patients at moderate risk of thrombosis include those with ASCVD without recent acute coronary syndrome (ACS) or percutaneous coronary intervention (PCI) and with high-risk features, those who have undergone elective PCI or those with medically managed non-ST-segment elevation ACS. Patients at high risk of thrombosis include those with ST-segment elevation myocardial infarction, stroke, stent thrombosis or PCI in ACS. The differentiation between upper GIB and lower GIB management comes from the American College of Gastroenterology guidelines on lower GIB[10]. **b**, Patients with a risk of thrombosis necessitating anticoagulation. Patients at low risk of thrombosis include those with atrial fibrillation (AF) (a $CHA_2DS_2$-VASc score of 2), isolated distal deep-vein thrombosis, subsegmental pulmonary embolism (PE), or venous thromboembolism (VTE) more than 6 months previously caused by a major provoking risk factor without any other risk factor. Patients at moderate risk of thrombosis include those with an intermediate–low risk PE, mechanical valves other than in the mitral position, high-risk AF, or VTE 1–6 months previously. Patients at high risk of thrombosis include those with high-risk PE, intermediate–high risk PE, a mechanical mitral valve, VTE in the past month, or thrombotic antiphospholipid syndrome in patients with acute bleeding. [a]In patients with active, clinically relevant GIB, treatment with antithrombotic agents should be paused until haemostasis stability and cessation of bleeding; clopidogrel is the P2Y purinoceptor 12 ($P2Y_{12}$) inhibitor of choice after a bleeding event. [b]In patients with upper GIB, aspirin treatment should be paused when used with $P2Y_{12}$ inhibitors. In patients with lower GIB, treatment with $P2Y_{12}$ inhibitors should be paused when used with aspirin. [c]In all patients with clinically relevant GIB, treatment with anticoagulants should be withheld. DAPT, dual antiplatelet therapy; OAC, oral anticoagulant; SAPT, single antiplatelet therapy.

## Box 2

# Knowledge gaps and research priorities

**Antithrombotic therapy in patients with malabsorption and alternative enteral dosage forms of antithrombotic agents**

- Randomized controlled trials (RCTs) evaluating the safety and efficacy of non-oral routes of administration of antithrombotic agents in patients with malabsorption.
- Correlation between the plasma levels of direct oral anticoagulants and clinical outcomes in patients with malabsorption.
- Correlation between platelet function tests and clinical outcomes in patients with malabsorption.
- Additional RCTs evaluating the safety and efficacy of antithrombotic agents in patients with bariatric or gastrointestinal resection surgery.
- RCTs (powered for clinical outcomes) on the effects of opioids and/or prokinetic agents in patients taking antiplatelet therapy.

**Gastrointestinal protection strategies in patients receiving antithrombotic agents**

- Prospective studies comparing the safety and efficacy (including the risk of gastrointestinal bleeding (GIB)) in patients receiving abbreviated dual antiplatelet therapy with second-generation versus third-generation drug-eluting stents.
- Risk scores that can assess the risk of GIB in patients receiving anticoagulation for atrial fibrillation.
- Additional prospective analyses assessing the association between use of proton pump inhibitors, including omeprazole and newer inhibitors, and clinical outcomes in those receiving clopidogrel.

**Management of antithrombotic therapy in patients with active GIB**

- RCTs comparing withholding versus continuing antiplatelet medications after haemostatic management in patients with acute GIB across the strata of thrombotic risk.
- RCTs comparing withholding versus continuing anticoagulant agents after haemostatic management in patients with acute GIB across the strata of thrombotic risk.

**Management of antithrombotic therapy in patients with recent GIB**

- RCTs comparing different times of resumption of antiplatelet agents on clinical outcomes after acute GIB across the strata of thrombotic risk.
- RCTs comparing different times of resumption of anticoagulant agents on clinical outcomes after acute GIB across the strata of thrombotic risk.
- Comparative studies on the safety and efficacy of procedural interventions versus anticoagulant agents in patients at risk of thrombotic events who have had a GIB event.
- RCTs comparing different combinations of antithrombotic agents on clinical outcomes after acute GIB in patients with different thrombotic risks.

### Antiplatelet therapy

Several guidelines recommend the resumption of aspirin therapy for secondary prevention on the same day if haemostasis is endoscopically confirmed, whereas resumption can be delayed in those receiving antiplatelet agents for primary prophylaxis[142,143]. Even when aspirin is given for secondary prevention, the balance between the risk of thrombosis and the risk of recurrent bleeding should be considered (Fig. 2). After hospitalization for diverticular haemorrhage, the continuation of aspirin for secondary prevention is also recommended after stabilization[10].

In patients with upper GIB and clear indications for DAPT, such as very recent stenting or myocardial infarction, an ESC consensus statement and European Society of Gastrointestinal Endoscopy guidelines recommend DAPT continuation if endoscopy shows low-risk features, whereas in the presence of active bleeding and high-risk endoscopic features, the guidelines recommend the short-term discontinuation of aspirin or DAPT based on bleeding severity and thrombotic risk, with at least one drug (usually aspirin) being reintroduced early and the second drug being withheld for up to 5 days after haemostasis[142,143]. A high dose of a PPI is recommended in all patients with a bleeding peptic ulcer[144]. In patients with a recently implanted stent and lower GIB while receiving DAPT, aspirin should be continued if possible, while the $P2Y_{12}$ inhibitor is discontinued for up to 5 days[10].

### Anticoagulant therapy

In patients with active GIB, anticoagulants should be discontinued[3,142,143]. Guideline recommendations for restarting anticoagulation vary from as soon as haemostasis is established[145], to within the first 3–7 days in patients with a high thromboembolic risk, to 7–15 days after haemostasis[143,145–148]. Resumption of anticoagulation within 7 days of a lower GIB event is also recommended in the American College of Gastroenterology guidelines[10] (Fig. 3).

## Patients with recent GIB (past 3 days)

### Antiplatelet therapy

Evidence favours the resumption of antiplatelet therapy after stabilization of GIB in patients with symptomatic vascular disease. In a placebo-controlled RCT in 156 patients treated with aspirin who presented with bleeding due to peptic ulcer disease, the resumption of low-dose aspirin after stabilization of bleeding was associated with markedly reduced mortality at 8 weeks (1.3% versus 12.9%; $P = 0.005$) at the cost of a non-significant increase in recurrent ulcer rebleeding at 30 days (10.3% versus 5.4%; $P = 0.25$)[149]. In patients at high risk of thrombosis (such as those with recent myocardial infarction, ischaemic stroke or PCI) who have low-risk endoscopic features (no identified source and a clean-based ulcer or no active bleeding)[150,151], the resumption of low-dose aspirin and $P2Y_{12}$ inhibitors (if indicated) is favoured shortly after achievement of haemostasis. The ESC guidelines suggest switching from ticagrelor or prasugrel to clopidogrel after GIB[142]. In patients with a lower risk of thrombosis (such as those with stable atherosclerotic cardiovascular disease) or a high risk of recurrent bleeding (active bleeding or a non-bleeding visible vessel)[150,151], delayed resumption of antiplatelet therapy is preferred. The length of delay depends on

the severity and source of the bleeding and should be determined in consultation with a multidisciplinary team.

### Anticoagulant therapy

Data from routine clinical practice indicate that OAC is withheld indefinitely in up to 50% of patients after acute major GIB[152–158]. Although the evidence is derived from observational studies with methodological issues and there is a risk that confounding factors might have influenced the resumption or discontinuation of anticoagulation, patients who resumed anticoagulation had substantially lower relative risks of thrombosis (40–70%) and death (40–50%), but up to a twofold higher relative risk of recurrent bleeding[156,157,159]. Bleeding risk, thrombosis risk[160], elapsed time from the bleeding[161] and functional status are among the most important factors that should influence the optimal time to restarting anticoagulation (Fig. 3). Given the absence of high-quality data, consideration of absolute risk, the potential for preventive procedures (for both bleeding and thrombosis), and the long-term risks of bleeding and thrombosis are crucial for determining net clinical benefit, using shared decision-making with input from a multidisciplinary team[79].

Long-term management of anticoagulation in these patients should incorporate periodic reviews to ensure appropriate drug and dose selection for the indication and patient characteristics (such as age, body weight and serum creatinine level) and to identify and manage modifiable bleeding risk factors, including those that develop during follow-up. A growing body of literature, although mostly in the form of retrospective or quality-improvement studies, suggests that stewardship programmes are coordinated, efficient and sustainable system-level initiatives that can optimize anticoagulant prescribing in hospitalized patients[162] (Box 1).

## Future directions and research gaps

Although newer antithrombotic agents and device technologies have reduced thrombotic events, several unmet needs still exist in relation to safety in patients with gastrointestinal comorbidities. Some modifications to make GIB less likely are under investigation[163]. Phospholipid-coated aspirin is associated with reduced gastrointestinal toxicity compared with plain aspirin[164]. In an RCT involving 204 healthy individuals, the rates of endoscopically assessed upper gastrointestinal injury was significantly higher in those assigned to 7 days of aspirin 325 mg daily as plain tablets than in those receiving aspirin 325 mg daily as phospholipid-coated tablets (42.2% versus 22.2%; $P = 0.0027$)[165]. Antibody-based reversal strategies are a new advance in antiplatelet therapy[166,167]. Inhaled aspirin, orodispersible ticagrelor tablets (which disperse upon contact with the moist mucosal surface) and the development of subcutaneous $P2Y_{12}$ inhibitors with a rapid onset of action are other strategies to improve the PK/PD properties of antiplatelet agents[168–170].

For anticoagulant therapy, several ongoing trials are evaluating the safety and efficacy of inhibitors of factor XI or factor XIa, which have the potential to reduce bleeding events[171–174]. Reconsidering older but less-studied agents with a potentially lower risk of bleeding might also be beneficial. Sulodexide, a purified glycosaminoglycan consisting of heparan sulfate and dermatan sulfate, has been shown to reduce the risk of venous and arterial thrombosis without an increase in bleeding risk, and could be a target for future studies in patients at high risk of bleeding[175] (Box 2).

## Conclusions

Maintaining the safety, without compromising the efficacy, of antithrombotic agents in patients with cardiovascular and gastrointestinal disorders is a common and important dilemma. Tailoring antithrombotic therapy and the management of patients with active or recent GIB and those at high risk of GIB often requires the involvement of a multidisciplinary team of experts.

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## Author contributions

A.H.T., H.K., M.A. and B.B. researched data for the article. All the authors discussed its content. A.H.T., P.S., L.O.-P., C.B., J.F., D.M.S., L.A.C., A.C., G.D.B., J.M.C., E.A.S., R.D.C., J.E.K., A.A., G.P. and B.B. wrote the manuscript. All the authors reviewed/edited it before submission.

## Competing interests

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**Review criteria** A diverse group of experts was convened (cardiologists, haematologists, gastroenterologists, bariatric surgeons, neurologists, internists, pharmacists and epidemiologists). A protocol outline was agreed ahead of the literature search. A series of comprehensive literature searches were performed to address several questions. These topics included pharmacokinetics and pharmacodynamics of antithrombotic agents with respect to gastrointestinal absorption impairment with motility problems, oral administration difficulties and tube feeding, and gastrointestinal resection surgery. Separate systematic search queries were prepared for strategies to reduce the risk of gastrointestinal bleeding (GIB) in patients receiving antithrombotic agents, including acid-suppressive medications and modifying antithrombotic regimens. Additional search queries were devised for the management of antithrombotic agents in patients with acute GIB, in those with recent GIB, and in those at high risk of GIB without active or recent bleeding (such as patients with inflammatory bowel disease or gastrointestinal cancer). The patient population, interventions, comparators, outcomes and type of study (PICOS) model was used as a search strategy tool. For efficacy or comparative effectiveness questions, randomized controlled trials and, where appropriate, pooled analyses from randomized controlled trials were prioritized. Medline with the PubMed interface was searched. In addition, a systematic search was conducted with respect to existing guidelines or major society recommendations related to the key questions. The date of the last search was 21 January 2023. Existing recent guidelines were referred to if they addressed a question, instead of duplicating them. Cost-effectiveness is not covered in this Review.

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